ITA0608 – Machine Learning Lab Practicals

EXP 05:

Write a program for Implementation of K-Nearest Neighbours (K-NN) in Python

Aim:

To implement the K-Nearest Neighbours (K-NN) algorithm in Python and classify new data samples using an appropriate dataset.

Algorithm:

- Choose the value of K (number of nearest neighbours).
- Calculate the distance between the test sample and all training samples.
- Select the K nearest neighbours based on the shortest distance.
- Count the class labels of these neighbours.
- Assign the class with the highest frequency to the test sample.

Program:

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.neighbors import KNeighborsClassifier

from sklearn.metrics import accuracy_score, classification_report

iris = load_iris()

X = iris.data

y = iris.target

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.3, random_state=42

)

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)

X_test = scaler.transform(X_test)

knn = KNeighborsClassifier(n_neighbors=3)

knn.fit(X_train, y_train)

y_pred = knn.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
```

```
print("Accuracy:", round(accuracy * 100, 2), "%")
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred, target_names=iris.target_names))
new_sample = [[5.1, 3.5, 1.4, 0.2]]
new_sample_scaled = scaler.transform(new_sample)
prediction = knn.predict(new_sample_scaled)
print("\nNew Sample:", new_sample[0])
print("Predicted Class:", iris.target_names[prediction][0])
```

Output:

```
Accuracy: 100.0 %

Classification Report:

              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        19
  versicolor       1.00      1.00      1.00        13
   virginica       1.00      1.00      1.00        13

    accuracy                           1.00        45
   macro avg       1.00      1.00      1.00        45
weighted avg       1.00      1.00      1.00        45


New Sample: [5.1, 3.5, 1.4, 0.2]
Predicted Class: setosa
```

Result:

The K-Nearest Neighbours (K-NN) algorithm was successfully implemented in Python using the Iris dataset.